Requirements for an Excavation and Lateral Support Plan
Building (Administration) Regulation 8(1)(bc)

Introduction

Bulk excavation, even if shallow, can be dangerous if not properly designed. Authorized persons, registered structural engineers (RSE) and registered geotechnical engineers (RGE) are reminded to take adequate precautions to ensure public safety whenever excavation is found necessary as part of the work. Reference should be made to PNAP APP-18 and PNAP APP-137 on details of monitoring requirements if excavation and lateral support (ELS) works, pile installation works and similar operations are proposed.

Criteria for Submission of ELS Plans

2. In general terms, ELS plans are required under Building (Administration) Regulation 8(1)(bc) to be submitted for approval where excavations of substantial depth are to be carried out, the collapse of which would cause serious consequences.

3. ELS plans will be required to be submitted to the Buildings Department (BD) for approval where the excavation works to be carried out are:

(a) deeper than 2.5 m and greater than 5 m in length; and

(b) liable to affect any road, building, structure, slope steeper than 30° or water main 75 mm in diameter or greater, the affected area being defined as within the 45° line up from the base of the excavation to the ground surface.

4. Where the above conditions apply, consent to commence ELS works would not be given if ELS plans, as prescribed in Building (Administration) Regulation 8(1)(bc), have not been submitted and approved.

5. Where ELS plans are required to be submitted for approval as stipulated above, an RSE would be required to prepare and sign the plans and structural design as well as the structural assessment report of the effects of the excavation and dewatering on adjoining structures. For ELS plans with excavation depth deeper than 4.5m, an RGE would be required to prepare and sign the supporting documentations, such as geotechnical assessment, geotechnical details and calculations, geotechnical reports, site investigation reports or ground investigation reports, which should accompany the ELS plans submission.

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6. It is permissible to have the ELS plans submitted in two stages, provided that the first stage submission demonstrates the feasibility of the entire scheme. Plans submitted in the first stage should show the pile wall system, for example sheet piles or a diaphragm wall, together with a strutting layout and the construction sequence. The supporting geotechnical documentation should also be provided in the first stage submission. Calculations submitted in the first stage should include realistic ground movement estimates for the entire works. The second stage submission typically would include the structural design and details of the lateral support system, including detailing of the walings and struts for each stage of excavation according to the approved first stage submission. To minimise the idling time on construction sites, earth-retaining elements such as sheet/pipe pile walls may be installed and, if applicable, initial stage ELS works[1] may be carried out concurrently with the foundation works. For further details, please refer to Appendix C of PNAP APP-18.

Global Factor Method

7. The Global Factor Method (GFM) has generally been used in the design of ELS works. In this method, unfactored soil strength parameters and loads are used in the stability checks. A minimum global factor of safety of 1.5 should be adopted for design against overturning/toe stability by using effective stress analysis with drained shear strength of soil. If total stress analysis with undrained shear strength of soil is used for design against overturning/toe stability[2], a minimum global factor of safety of 2.0 should be adopted. Guidance and requirements are given in the "Notes on Design of Excavation and Lateral Support Works" in **Appendix A**.

Partial Factor Method

8. With the publication of GEO Publication No. 1/2023 — Deep Excavation Design and Construction by the Geotechnical Engineering Office of the Civil Engineering and Development Department in 2023, Partial Factor Method (PFM) could be used, as an alternative, for analysis and design of ELS works. In this method, soil structure interaction analysis is carried out for both the serviceability limit state and ultimate limit state conditions. Soil parameters and loads should be applied with partial factors in the design of ELS works. This alternative method should also conform to the guidance and requirements given in the "Notes on Design of Excavation and Lateral Support Works" in **Appendix A**.

Submission Requirements

9. In all ELS plans submissions, the design method used should be stated on the plans and calculations. The submission procedures for ELS works adopting either GFM or PFM will be the same. However, ELS works design based on a mixture of both methods is not allowed.

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[1] The initial stage ELS works refer to the excavation down to the bottom of the first layer of struts but not exceeding 1.5 m below the existing ground level and the erection of the first layer of waling and struts as shown in the approved plans.

[2] The factor of safety against loss of moment equilibrium of the embedded wall should be applied on the passive earth pressure. Water pressure should not be factored.

10. Computer programs to be used to support the design proposals should be submitted to BD for prior acceptance. The general requirements for prior acceptance of computer programs are given in PNAP ADM-6. Even if a computer program is applicable to both design methods, separate application for prior acceptance of the computer program for each design method is required.

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Appendix A
(PNAP APP-57)

Notes on Design of Excavation and Lateral Support Works

1. Introduction

1.1 This document provides technical guidance and requirements for the design of excavation and lateral support (ELS) works in Hong Kong. Those given in Section 2 are general guidance and requirements for ELS works and are applicable to both the Global Factor Method (GFM) and Partial Factor Method (PFM). Those given in Section 3 are applicable to the PFM only.

2. General Guidance and Requirements (APPLICABLE TO BOTH GFM AND PFM)

2.1 Serviceability Limit State (SLS) Checks

2.1.1 Where necessary, SLS checks should be carried out to assess the impacts on adjoining buildings, structures, services and land.

2.1.2 For the assessment of pile wall deflections and associated ground movements at different stages of ELS works including dewatering and excavation, the guidance given in GEO Publication No. 1/2023 — Deep Excavation Design and Construction should be followed.

2.2 Soil Stiffness

2.2.1 In applying soil stiffness from correlations with SPT 'N' values based on reliable published case histories, account should be taken of the soil type and the computer program used in the back analyses for deriving the correlation.

2.3 Base Heave and Hydraulic Failure

2.3.1 The guidance given in Section 6.2.3 and 6.2.4 of GEO Publication No. 1/2023 should be followed and a minimum factor of safety of 1.5 against base heave and hydraulic failure should be used.

2.3.2 In the design, account should be taken of the uncertainties in ground conditions (e.g. layered soil with heterogeneity in permeability) and the applicability of the design charts for the particular site conditions.

2.4 Design Report and Design Review

2.4.1 The design report should be certified by the RSE and RGE. It should state that:

(a) the analyses demonstrate that there is an adequate safety margin against collapse of the excavation, and

/(b) …

(b) there will be no unacceptable impacts due to the ELS works provided that the design assumptions are verified and found to be valid and the works are carried out as specified during construction.

2.4.2 The design report should indicate the design assumptions required to be reviewed and verified during construction, the range of computation results for the key parameters requiring monitoring and checking, and the critical working procedures requiring inspection and certification during construction.

2.4.3 A copy of the design report should be made available to the qualified site supervision personnel responsible.

2.4.4 Monitoring should be carried out to review and check the performance of the design during construction. As a minimum, ground settlement/heave and piezometric readings should be monitored to provide data for design review, risk management/control and performance review. Where there is significant risk to life and property in the event of a failure, monitoring of pile wall deflections and associated ground movements, water leakage and ground loss through or beneath the wall, as well as the change to the flow of groundwater should also be carried out.

3. Guidance and Requirements (APPLICABLE TO PFM ONLY)

3.1 Scope

3.1.1 PFM can be used for ELS works including those with pile walls socketed in rock or embedded in soil.

3.2 Design Parameters

3.2.1 The partial factors and surcharge loads recommended in GEO Publication No. 1/2023, together with its guidance on site investigation, ground modelling, selection of geotechnical parameters and design water pressures, should be adopted except that the partial load factor of 1.5 for surcharge load should be adopted instead of 1.3.

3.3 Structural Design

3.3.1 PFM should be used for structural design. A factor of 1.4 should be applied to the forces and bending moments obtained from the SLS calculations carried out in accordance with the GEO Publication No.1/2023.

3.3.2 Additional structural calculations using the partial factors given in the relevant structural design codes may be required to check the structural members.

3.4 Ultimate Limit State (ULS) Checks

3.4.1 ULS calculations should be carried out using limit equilibrium method or soil structure interaction analysis to ensure stability with an adequate safety margin.

/3.4.2 …

3.4.2 When ULS checks are based on the recommendations given in this Appendix, there is no need to carry out separate checks against moment instability (i.e. overturning and kick-out failures) using a global factor of safety of 1.5 (for effective stress analysis) or 2.0 (for total stress analysis), as required in GFM.

3.5 Sensitivity Analyses in ULS Checks

3.5.1 Under the GEO Publication No.1/2023, the limit equilibrium method may be used to determine the minimum wall embedment depth required to ensure an adequate safety margin against moment instability. Alternatively, soil structure interaction analyses method may be carried out.

3.5.2 In soil structure interaction analyses, there could be a rapid increase in the maximum pile wall deflection/strut loads upon a small reduction in the wall embedment depth, reflecting the sensitivity of the design to small variations in wall embedment. Hence a suitable value of design wall embedment depth should be selected to take into account the results of sensitivity analysis and the construction tolerance that can be achieved under the construction control and supervision regime imposed.

3.6 Design of ELS Works Socketed in Rock

3.6.1 For the design principles and instability checks of rock sockets, reference should be made to PFM for the design of rock-socketed cantilever walls as outlined in Chapter 7 of GEO Publication No. 1/2023.

(Rev. 7/2026)